

# Zhuoyang Chen

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## Education

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### University of Michigan

Ann Arbor, MI, United States

B.E. in Computer Engineering

- GPA: 3.85/4.0
- Coursework: Algorithm and Data Structure (A), Automatic Control (A+), Digital Control System (A), Computer Vision (A), Algorithmic Robotics (A), Reinforcement Learning Theory (A)

### Shanghai Jiao Tong University

Shanghai, China

B.E. in Mechanical Engineering

- GPA: 3.63/4.0
- Coursework: Honors Calculus II & III (A, A+), Linear Algebra (A), Circuits (A), Dynamics and Vibration (A)

## Publications and Manuscripts

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**Z. Chen**, X. Wang, S. Revzen., “Walking on Rough Terrain with Any Number of Legs.”

*Bioinspiration & Biomimetics*. 2026. <https://iopscience.iop.org/article/10.1088/1748-3190/ae9261/meta>

J. Cao\*, **Z. Chen\***, et al, “Unrolling the Action Manifold: Visuomotor Policy Learning via Recursive Cascades” *Conference on Robot Learning*. 2026 (Under Review).

## Research Experience

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### BIRDS Lab, University of Michigan

Nov 2024 – May 2026

Undergraduate Researcher | Advisor: Dr. Shai Revzen

- Proposed and implemented a **bio-inspired cascaded, hierarchical finite state machine (FSM) controller** for a compliant-joint, modular legged robot, enabling stable walking on rough terrain with any number of legs and minimal sensor input, at a **walking speed as fast as 4 times the body length** in simulation.
- Designed a bio-inspired modular robot (Autodesk Inventor/Onshape) with compliant joints and integrated a contact-sensing system to support gait state transitions.
- Validated the distributed FSM control architecture in MuJoCo simulator and deployed an open-loop version to a physical robot through an embedded ESP32 control system.

### ROAHM Lab, University of Michigan

May 2025 – Present

Research Assistant (UM-SURE) | Advisor: Dr. Ram Vasudevan, Dr. Yue Hu

- Extended Tidybot++ (holonomic powered-caster base + 7-DOF manipulator) by building an end-to-end ROS2 stack with MoveIt2 in Gazebo and Isaac Sim simulators, and on real hardware, accompanied with an Isaac Lab reinforcement learning training environment  
[https://github.com/roahmlab/tidybot\\_platform.git](https://github.com/roahmlab/tidybot_platform.git)
- Implemented a full pipeline on top of the extended Tidybot++ platform for teleoperation, dataset collection, and imitation learning policy training/inference. Trained and deployed the Diffusion Policy in simulation and on hardware; fine-tuned and deployed OpenVLA and  $\pi_0$ ,  $\pi_{0.5}$  on hardware.
- Contributed to a Long-Horizon Mobile Manipulation project by extending the Tidybot++ ROS2 platform and generating expert mobile manipulation trajectory using MoMaGen, and evaluating sampled base positions.

### University of Hong Kong

March 2026 – Present

Research Assistant | Advisor: Dr. Andrew Luo

- Proposed and implemented a coarse-to-fine trajectory generation method for robotic manipulation by hierarchical and recursive action instantiation. The policy achieved **doubled performance than diffusion policy and ACT** on RoboTwin2.0, beating  $\pi_0$  and *RDT*, **with only  $\sim 19M$  parameters and 2ms/waypoint generation rate**.
- Evaluated the policy on RoboTwin2.0, ManiSkill, and on hardware; co-authored an academic paper (Under

review at CoRL 2026). <https://sagecao1125.github.io/RCP-Site/>

## Shanghai Institute for Physical Artificial Intelligence and Robotics (PAIR)

May 2026 – Present

Research Intern | Advisor: Dr. Yutong Ban

- Proposed and implemented a **dual-residual neural-augmented simulator** for dexterous manipulation sim2real transfer. The neural-augmented simulator consists of an object-side residual external impulse module and a hand-side residual action module.
- Evaluated the dual-residual neural-augmented simulator on fingertip object-rotation and in-hand object-reorientation policies trained in MuJoCo in Genesis, **closing**  $\sim 60\%$  of **Sim2Sim** gap.

## Teaching Experience

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### Intro to Engineering

May 2024 – Aug 2024

Teaching Assistant, UM-SJTU Joint Institute

- Led labs and office hours on mechatronic prototyping with Arduino and mechanical CAD with SolidWorks.
- Mentored student teams to design and manufacture a rover (paper-based truss structure) capable of transporting loads.
- Created homework on mechatronic system design and supported course events (game day, symposium, design expo).

### Intro to Signals and Systems

Jan 2026 – Present

Grader, University of Michigan

- Graded homework and pre/post lab assignments, and responded to students' regrade requests.

## Leadership Experience

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### Student Science and Technology Innovation Association (SSTIA)

May 2024 – Aug 2024

Co-Leader, UM-SJTU Joint Institute, Shanghai Jiao Tong University

- Organized workshops or sharing STEM skills like doing CAD with SolidWorks, academic writing with LaTeX, and hosted guest speaker for seminars sharing entrepreneurship and academic experiences.
- Hosted an annually university-wide Liming-Cup Undergraduate Robotics Competition with more than 30 participating teams and designed a prototype mobile manipulation robot for participants' reference.
- Organized info session and training workshops for The University Physics Competition (UPC) and The Mathematical Contest in Modeling (MCM).

## Selected Honors and Awards

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### University of Michigan

Ann Arbor, MI

- 2025-2026 EECS Outstanding Achievement Award (only 3 recipients per year)

Mar 2026

- University Honors, Dean's List

Dec 2024, May 2025, Dec 2025

### Shanghai Jiao Tong University

Shanghai, China

- Shanghai Jiao Tong University Excellent Graduate

Aug 2026 2026

- Excellent Undergraduate Scholarship (Top 10%)

Dec 2023, Dec 2024

- John Wu and Jane Sun Sunshine Scholarship (Top 6%)

Dec 2023

- Second Prize, Shanghai Undergraduate Engineering Practice And Innovation Ability Competition

Oct 2023

## Professional Skills

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**Programming:** C/C++, Python, Pytorch, MATLAB

**Robotics/Simulation:** ROS2, Isaac Sim/Lab, MuJoCo, Gazebo

**Embedded/Hardware Tools:** ESP32, STM32 (CubeMX/CubeIDE), Raspberry Pi, Nvidia Jetson; PCB design with KiCad

**Mechanical CAD:** Autodesk Inventor, SolidWorks, Onshape